Yujie Cen

cenyjie@gmail.com Vienna, Austria diroccen.github.io github.com/dirocCen orcid.org/0000-0002-2111-0191
scholar.google.com/citations?user=IuwWrz0AAAAJ

Research Profile

Computational materials scientist working on first-principles simulations, machine-learning interatomic potentials, and symmetry-aware methods for thermal transport and low-dimensional materials.

Education

PhD in Materials Chemistry — 07/2023 — 10/2026, Vienna, Austria
Vienna University of Technology (TU Wien) · degree expected October 2026
Supervisor: Georg K. H. Madsen · Co-supervisor: Jesús Carrete Montaña

MSc in Theoretical Physics — 09/2020 — 07/2023, Guangzhou, China
South China University of Technology
Supervisor: Xiaobao Yang

BSc in Physics — 09/2016 — 07/2020, Guangzhou, China
South China University of Technology

Research Experience

Active learning and machine-learning interatomic potentials — 2025 — 2026, TU Wien, Vienna
arXiv:2609.02397

- Generated first-principles reference data with VASP and fine-tuned a neuroevolution potential for a TMD nanotube through two rounds of active learning.
- Built reproducible Python workflows covering structure generation, training, validation and GPU-accelerated molecular dynamics in GPUMD.
- Validated the potential against physics-based observables — structural stability, phonon properties and finite-temperature transport — not force and energy errors alone.

Physics-aware machine learning for atomistic transport — 2024 — 2025, TU Wien, Vienna
npj Computational Materials, 2026

- Coupled an Allegro equivariant machine-learning potential with a symmetry-adapted, mode-resolved atomistic Green's-function framework for WS_2–MoS_2 nanotubes.
- Showed that symmetry-breaking defects open transmission channels the symmetric structure forbids, so the more disordered configuration conducts about 60% better.
- Confirmed independently by Green–Kubo molecular dynamics: 9.8 ± 0.8 against 6.1 ± 0.6 W m^{-1} K^{-1}.

High-throughput materials discovery in alloy configuration space — 2021 — 2023, South China Univ. of Technology, Guangzhou
Physical Review Materials, 2022

- Partitioned cluster-expansion correlation space by symmetry to make convex-hull optimisation tractable at high throughput.
- Recovered every known ground state across five alloy systems and predicted previously unreported lower-energy configurations.

Open-Source Software

Pulgon-tools — Python toolkit for symmetry analysis of quasi-1D materials — 2023 — 2026
Lead developer · Computer Physics Communications, 2026

- First openly available package able to detect the line group of an arbitrary quasi-1D atomistic structure, covering all 13 line-group families; three-dimensional space-group detectors return only a low-symmetry result for nanotubes and nanowires.
- Implements two-stage line-group detection, irreducible representations and character tables labelled by physical quantum numbers, symmetry-aware structure generation, and constrained force-constant correction enforcing translational and rotational invariance; shipped with both a command-line interface and a Python API.

github.com/pulgon-project/pulgon_tools

Publications

Symmetry-based modal analysis of heat transport in molecular dynamics of quasi-1D systems 09/2026
arXiv:2609.02397 (preprint)
Cen, Y.-J., Wieser, S., Madsen, G. K. H. & Carrete, J.
arxiv.org/abs/2609.02397

Pulgon-tools: a toolkit for analysing and harnessing symmetries in quasi-1D systems 08/2026
Computer Physics Communications
Cen, Y.-J., Wieser, S., Madsen, G. K. H. & Carrete, J.
doi.org/10.1016/j.cpc.2026.110351

Ab-initio heat transport in defect-laden quasi-1D systems from a symmetry-adapted perspective 01/2026
npj Computational Materials 12(1), 19
Cen, Y.-J., Wieser, S., Madsen, G. K. H. & Carrete, J.
nature.com/articles/s41524-025-01866-1

Accelerating first-principles molecular-dynamics thermal conductivity calculations for complex systems 09/2025
Journal of Chemical Theory and Computation
Wieser, S., **Cen, Y.-J.**, Madsen, G. K. H. & Carrete, J.
doi.org/10.1021/acs.jctc.5c01525

Residual entropy of ice: a study based on transfer matrices 11/2024
Physical Review B 110(19), 195414
Li, D. Z., **Cen, Y.-J.**, Wang, X. & Yang, X. B.
doi.org/10.1103/PhysRevB.110.195414

Determining ground states of alloys by a symmetry-based classification 05/2022
Physical Review Materials 6(5), L050801
Cen, Y.-J., He, C. C., Qiu, S. B., Zhao, Y. J. & Yang, X. B.
doi.org/10.1103/PhysRevMaterials.6.L050801

Presentations

Ab-initio heat transport in defect-laden quasi-1D systems from a symmetry-adapted perspective
Oral presentation · DPG Meeting 2025

Research Methods & Computational Tools

Machine-learning interatomic potentials
NEP, MACE, Allegro, NequIP; active learning; model fine-tuning; equivariant graph neural networks; PyTorch

First-principles & molecular dynamics
DFT (VASP); molecular dynamics (GPUMD, LAMMPS); ASE; Phonopy; lattice dynamics; phonon and thermal-transport calculations

Research software & HPC
Python; NumPy/SciPy; package and CLI development; reproducible workflows; testing; Git; Bash; Linux; SLURM; GPU clusters

Symmetry & group theory
Space and line groups; irreducible representations; symmetry-adapted bases; constrained force-constant correction

Languages

Chinese
native

English
fluent